

Literature Benchmark Brief: Doherty_2005_TPWRS

Title: A new approach to quantify reserve demand in systems with significant wind penetration

Authors: R. Doherty, M. O'Malley (2005)

Venue: IEEE Transactions on Power Systems | **DOI:** 10.1109/TPWRS.2005.846206

Cluster: Cluster D: Probabilistic Sizing & Reserve

Core Theoretical Mechanism & Model Formulation

Statistical convolution of wind forecast error distributions for spinning reserve sizing

Empirical Scale & Experimental Dataset

Irish transmission grid operational model

Relevance to IEEE TSTE & Sustainable Energy Systems

Pioneered explicit reserve margin sizing based on wind forecast error percentiles

Identified Gaps & Operational Vulnerabilities

Assumed static Gaussian error distributions across all operational regimes

Role in Manuscript Defense & Reviewer #2 Proof

Cited in Section I and II to explain asymmetric shortage penalty rationale